



Cardiovascular pharmacology

Analysis of anandamide- and lysophosphatidylinositol-induced inhibition of the vasopressor responses produced by sympathetic stimulation or noradrenaline in pithed rats

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ABSTRACT

The endocannabinoid system exhibits multiple functions in cardiovascular regulation mainly by cannabinoid (CB₁ and CB₂) receptors, vanilloid TRPV1 receptors and, probably, by the orphan G protein-coupled receptor 55 (GPR55). Hence, the role of these receptors was investigated in Wistar pithed rats on anandamide- and lysophosphatidylinositol (LPI)-induced inhibition of the vasopressor responses induced by preganglionic (T₇–T₉) stimulation of the vasopressor sympathetic outflow or i.v. bolus injections of noradrenaline. The corresponding frequency- and dose-dependent vasopressor responses were analyzed before and during i.v. continuous infusions of anandamide (CB₁, CB₂, TRPV1 and GPR55), JWH-015 (CB₂) and LPI (GPR55) in animals receiving (i.v.) the antagonists NIDA41020 (CB₁), AM630 (CB₂), capsazepine (TRPV1) and/or cannabidiol (GPR55). Anandamide (0.1–3.1 µg/kg min) inhibited the vasopressor responses by electrical stimulation, but not those by noradrenaline; while LPI (5.6–10 µg/kg min) inhibited both responses. In contrast, JWH-015 (5.6–10 µg/kg min) failed to induce sympatho-inhibition. Anandamide-induced sympatho-inhibition was: (i) dose-dependently blocked by 31 and 100 µg/kg NIDA41020; (ii) slightly blocked by 310 µg/kg AM630 or 31 µg/kg cannabidiol; and (iii) unaffected by 310 µg/kg capsazepine. Moreover, LPI-induced inhibition of both vasopressor responses was blocked and abolished by 10 and 31 µg/kg cannabidiol, respectively, and weakly blocked by 100 µg/kg NIDA41020. Thus, the sympatho-inhibition by anandamide is primarily mediated by cannabinoid CB₁ and, minimally, by cannabidiol-sensitive receptors. In contrast, LPI-induced inhibition of both responses seems to be mainly mediated by postjunctional cannabidiol-sensitive (presumably endothelial GPR55) receptors.

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1. Introduction

The endocannabinoid system exhibits multiple biological functions including, among others, pain modulation, cell proliferation, cardiovascular homeostasis as well as appetite and body weight control (Di Marzo, 2011; Kirkham, 2005; Walker and Huang, 2002). These functions involve the activation of cannabinoid and non-cannabinoid receptors. With the conjunction of structural, transductional and operational criteria, cannabinoid (CB) receptors have been classified into CB₁ and CB₂ types (Matsuda et al., 1990; Munro et al., 1993). These receptors are coupled to G_{i/o}-proteins and their signal transduction mechanisms include inhibition of

adenylyl cyclase, activation of mitogen-activated protein kinase and, in the case of the cannabinoid CB₁ type receptor, regulation of Ca²⁺ and K⁺ channels (Pertwee, 1997). While cannabinoid CB₁ receptors modulate neurotransmitter release, cannabinoid CB₂ receptors are mainly present on immune cells (Pertwee and Ross, 2002; Sugiura and Waku, 2002).

The endogenously-produced cannabinoids include the eicosanoids anandamide, 2-arachidonoylglycerol, 2-arachidonoylglycerol ether, virhodamine and N-arachidonoyl-dopamine (De Petrocellis et al., 2004). Interestingly, anandamide induces local and systemic effects on the cardiovascular system through a wide variety of receptors including, among others: cannabinoid CB₁/CB₂ receptors (Pacher et al., 2008), non-cannabinoid CB₁/CB₂ receptors (Herradón et al., 2007) and vanilloid TRPV1 receptors (Poblete et al., 2005). Likewise, anandamide seems to interact with GPR55 (see Table 1) and causes vasorelaxation of the pulmonary artery (Baranowska-Kuczek et al., 2012); this vasorelaxing mechanism

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Table 1

Binding affinity constants (pK_i) of several ligands for CB₁, CB₂, TRPV1 and GPR55 receptors, except for α , β and γ , which stand for pEC_{50} , PK_B and pIC_{50} , respectively, determined in transfected cells.

Compound	CB ₁	CB ₂	TRPV1	GPR55
Agonist				
Anandamide	~7.05 ^a	~6.4 ^a	5.78 ^b (r)	7.7 ^{a,c}
JWH-015	~6.4 ^a	~7.8 ^a	N.D.	< 4.5 ^{a,d}
Lysophosphatidylinositol	< 4.5 ^{a,d}	N.D.	N.D.	7.31 ^{a,e}
Antagonist				
NIDA42020	~8.4 ^f	6.0 ^g	N.D.	N.D.
AM630	~5.2 ^h	~7.5 ^h	N.D.	N.D.
Capsazepine	N.D.	N.D.	~5.8 ^b (r)	N.D.
Cannabidiol	~5.3 ⁱ	~5.4 ⁱ	N.D.	6.3 ^{r,c}

(r) Interaction of anandamide with vanilloid receptors in CHO cells transfected with rVR1 (TRPV1) receptors; N.D. not determined; ~stands for “approximately”.
Data taken from.

^a Showalter et al. (1996).

^b Ross et al. (2001).

^c Ryberg et al. (2007).

^d Kapur et al. (2009).

^e Oka et al. (2007).

^f Katoch-Rouse et al. (2003).

^g Donohue et al. (2006).

^h Ross et al. (1999).

ⁱ Thomas et al. (2007).

could be related with endothelial GPR55 (Waldeck-Weiermair et al., 2008). Although the GPR55 may correspond to the novel cannabinoid CB₃ receptor, the high activation of these receptors by lysophosphatidylinositol (LPI; Oka et al., 2007, 2009) raises controversy as to whether it may be considered a cannabinoid receptor.

A rapid i.v. bolus injection of anandamide produces a triphasic blood pressure response in anaesthetised rats consisting of: (i) an initial transient bradycardic and hypotensive phase (abolished in pithed rats) due to stimulation of vagal afferents via vanilloid TRPV1 receptors; (ii) a rapid pressor response produced by an increase in cardiac contractility and systemic vasoconstriction; and (iii) a late, long-lasting hypotensive response mainly mediated by negative cardiac inotropy, systemic vasodilatation and sympatho-inhibition (Bonz et al., 2003; Malinowska et al., 2012; Zakrzewska et al., 2010). This sympatho-inhibition induced by anandamide and related agonists has been reported to involve activation of cannabinoid CB₁ receptors (Ishac et al., 1996; Malinowska et al., 1997), a response absent in cannabinoid CB₁-knockout mice (Ledent et al., 1999). However, the possible role of cannabinoid CB₂, GPR55 and vanilloid TRPV1 receptors in this sympatho-inhibition has not been reported. Hence, this study sets out to investigate by pharmacological means in pithed rats the specific role of cannabinoid (CB₁ and CB₂), GPR55 and vanilloid TRPV1 receptors in anandamide- and LPI-induced inhibition of the vasopressor responses induced by: (i) preganglionic (T₇–T₉) stimulation of the sympathetic vasopressor outflow; and (ii) i.v. bolus injections of exogenous noradrenaline.

2. Methods

2.1. Animals

Male Wistar normotensive rats (250–300 g) were maintained at a 12/12-h light-dark cycle (light beginning at 07:00 h) and kept in a special room at 22 ± 2 °C and 50% humidity, with food and water freely available in their home cages. All experimental protocols in this study were approved by our Institutional Ethics

Committee (CICUAL-Cinvestav), in accordance with the guide for the Care and Use of Laboratory Animals in the U.S.A.

2.2. General methods

Experiments were carried out in a total of 234 rats. After quick anaesthesia with diethyl ether and cannulation of the trachea, the rats were pithed by inserting a stainless steel rod through the orbit and foramen magnum into the vertebral foramen, as previously established by Shipley and Tilden (1947). Then, the animals were artificially ventilated with room air using a model 7025 Ugo Basile pump (56 strokes/min; stroke volume: 20 ml/kg) (Kleinman and Radford, 1964). After bilateral vagotomy, catheters were placed in the left and right femoral and jugular veins, for the infusion of the agonists by a WPI model sp100i pump (World Precision Instruments Inc., Sarasota, FL, U.S.A.) and bolus injections of the antagonists or vehicles; and the left carotid artery, connected to a Grass pressure transducer (P23 XL), for the recording of blood pressure. Heart rate was measured with a tachograph (7P4, Grass Instrument Co., Quincy, MA, U.S.A.) triggered from the blood pressure signal. Both blood pressure and heart rate were recorded simultaneously by a model 7 Grass polygraph (Grass Instrument Co., Quincy, MA, U.S.A.). At this point, the 234 rats were divided into two main sets, so that the effects produced by i.v. continuous infusions of dimethyl sulfoxide 0.5% (DMSO 0.5%), LPI vehicle (0.04 ml chloroform:methanol:water 70:27:3 + 0.96 ml saline) as well as the agonists anandamide, JWH-015 and LPI could be investigated on the vasopressor responses induced by either: (i) selective preganglionic (T₇–T₉) stimulation of the vasopressor sympathetic outflow (set 1; $n=180$); or (ii) i.v. bolus injections of exogenous noradrenaline (set 2; $n=54$). The vasopressor stimulus–response curves (S–R curves) and dose–response curves (D–R curves) elicited by, respectively, sympathetic stimulation and exogenous noradrenaline were completed in about 30 min without changes in resting diastolic blood pressure or heart rate. Moreover, the vasopressor sympathetic stimuli and the noradrenaline injections were given using a sequential schedule, in 0.5 log unit increments at 3- to 5-min intervals. The body temperature of each pithed rat was maintained at 37 °C by a lamp and monitored with a rectal thermometer.

2.3. Experimental protocols

2.3.1. Electrical stimulation of the vasopressor sympathetic outflow

In the first set of rats ($n=180$), the pithing rod was replaced by an electrode enamelled except for a 1-cm length 9 cm from the tip, so that the uncovered segment was situated at T₇–T₉ in the spinal cord to allow selective preganglionic stimulation of the thoracic sympathetic nerves supplying the systemic vasculature, i.e. the vasopressor sympathetic outflow; an indifferent electrode was placed dorsally (Gillespie et al., 1970; Villalón et al., 1995, 1998). Prior to electrical stimulation, all animals received gallamine (25 mg/kg, i.v.) to avoid electrically-induced muscular twitching. After 10 min, the animals were divided into two main sets ($n=156$ and 24). The first set of animals ($n=156$) was systematically pretreated with desipramine (50 µg/kg, i.v.) before each S–R curve. Under these conditions, the vasopressor responses to lower stimulation frequencies are greater than those elicited without desipramine (Ruiz-Salinas et al., 2013; Villalón et al., 1995, 1998). The second set of animals ($n=24$) did not receive desipramine and was performed for comparative purposes. After stable haemodynamic conditions for at least 30 min, baseline values of diastolic blood pressure and heart rate were determined in both sets of animals.

The preganglionic vasopressor sympathetic outflow was then stimulated in both sets to elicit vasopressor responses by applying

10-s trains (2 ms monophasic rectangular pulses, 50 V), at increasing frequencies (0.03, 0.1, 0.3, 1 and 3 Hz) using a sequential schedule. When diastolic blood pressure had returned to baseline levels, the next frequency was applied. This procedure was performed systematically until the S–R curve had been completed (about 30 min). The body temperature of each pithed rat was maintained at 37 °C by a lamp and monitored with a rectal thermometer. At this point, the first set of animals was divided into four treatment groups ($n=42$, 42, 24 and 48; **Protocol 1**), while the second set was divided into four groups ($n=6$ each; **Protocol 2**).

2.3.1.1. Protocol 1. The first group ($n=42$) was subdivided into seven subgroups ($n=6$ each) that received an i.v. continuous infusion of: (i) DMSO 0.5% (vehicle of anandamide and JWH-015, two times; 0.02 ml/min); (ii) LPI vehicle (0.04 ml chloroform: methanol:water 70:27:3 + 0.96 ml saline; twice; 0.02 ml/min); (iii) anandamide (0.1 and 0.31 $\mu\text{g/kg min}$); (iv) anandamide (1 and 3.1 $\mu\text{g/kg min}$); (v) LPI (1 and 3.1 $\mu\text{g/kg min}$); (vi) LPI (5.6 and 10 $\mu\text{g/kg min}$); and (vii) JWH-015 (5.6 and 10 $\mu\text{g/kg min}$). Ten minutes later, an S–R curve was elicited again during the above infusions to analyse their effects on the sympathetically-induced vasopressor responses. The intervals between the different infusions ranged between 10 and 15 min, as in each case we waited until diastolic blood pressure returned to baseline values.

The second group ($n=42$) was subdivided into seven subgroups ($n=6$ each) that received an i.v. bolus injection of: (i) DMSO 0.5% (vehicle of NIDA41020 and AM630; 1 ml/kg); (ii) DMSO 2% (vehicle of capsazepine and cannabidiol; 1 ml/kg); (iii) NIDA41020 (100 $\mu\text{g/kg}$); (iv) AM630 (180 $\mu\text{g/kg}$); (v) AM630 (310 $\mu\text{g/kg}$); (vi) capsazepine (310 $\mu\text{g/kg}$); and (vii) cannabidiol (31 $\mu\text{g/kg}$). Twenty minutes later, an S–R curve was elicited again, as described above, to analyse their effects per se on the sympathetically-induced vasopressor responses.

The third group ($n=24$), subdivided into four subgroups ($n=6$ each), received an i.v. bolus injection of: (i) DMSO 2% (1 ml/kg); (ii) NIDA41020 (100 $\mu\text{g/kg}$); (iii) cannabidiol (10 $\mu\text{g/kg}$); and (iv) cannabidiol (31 $\mu\text{g/kg}$). Ten minutes later, all subgroups received an i.v. continuous infusion of LPI (10 $\mu\text{g/kg min}$). After 10 min, an S–R curve was again constructed during the infusion of LPI as described above, to analyse the effects of the aforementioned compounds on LPI-induced inhibition of the electrically-induced vasopressor responses.

The fourth group ($n=48$) was subdivided into eight subgroups ($n=6$ each) that received an i.v. bolus injection of, respectively: (i) DMSO 0.5% (1 ml/kg); (ii) DMSO 2% (1 ml/kg); (iii) NIDA41020 (31 $\mu\text{g/kg}$); (iv) NIDA41020 (100 $\mu\text{g/kg}$); (v) AM630 (180 $\mu\text{g/kg}$); (vi) AM630 (310 $\mu\text{g/kg}$); (vii) capsazepine (310 $\mu\text{g/kg}$); and (viii) cannabidiol (31 $\mu\text{g/kg}$). Ten minutes later, all subgroups received an i.v. continuous infusion of anandamide (1 $\mu\text{g/kg min}$). Ten minutes after starting the continuous infusion of anandamide, an S–R curve was elicited again, as described above.

2.3.1.2. Protocol 2. The four groups of this protocol ($n=6$ each) received i.v. continuous infusions of: (i) DMSO 0.5% (0.02 ml/min; twice); (ii) LPI vehicle (0.02 ml/min; twice); (iii) anandamide (1 and 3.1 $\mu\text{g/kg min}$); and (iv) LPI (5.6 and 10 $\mu\text{g/kg min}$). Ten minutes after starting these infusions, an S–R curve was elicited (see above) during the infusion of every compound to analyse their effects on the electrically-induced vasopressor responses without desipramine. These experiments were performed to ascertain whether stimulation of inhibitory receptors by anandamide and LPI is also operative in the absence of desipramine, as previously reported for serotonin (Villalón et al., 1995) and quinpirole (Ruiz-Salinas et al., 2013).

2.3.2. Administration of exogenous noradrenaline

The second set of rats ($n=54$) were prepared as described above, but the pithing rod was left throughout the experiment. After determining baseline values of diastolic blood pressure and heart rate, vasopressor responses were elicited by administering i. v. bolus injections of exogenous noradrenaline at increasing doses (0.03, 0.1, 0.3, 1 and 3 $\mu\text{g/kg}$). When diastolic blood pressure had returned to baseline levels, the next dose was applied. This procedure was performed until the D–R curve had been completed (about 30 min). Subsequently, the animals were divided into two groups ($n=24$ and 30, respectively).

The first group ($n=24$) was subdivided into four subgroups ($n=6$ each) which received i.v. continuous infusions of, respectively: (i) DMSO 0.5% (two times; 0.02 ml/min); (ii) LPI vehicle (two times; 0.02 ml/min); (iii) anandamide (1 and 3.1 $\mu\text{g/kg min}$); and (iv) LPI (5.6 and 10 $\mu\text{g/kg min}$). Ten minutes after starting these infusions, a D–R curve for noradrenaline was elicited again during the above infusions.

The second group ($n=30$) was subdivided into five subgroups ($n=6$ each). Four subgroups received an i.v. bolus injection of: (i) DMSO 2% (1 ml/kg); (ii) NIDA41020 (100 $\mu\text{g/kg}$); (iii) cannabidiol (10 $\mu\text{g/kg}$); and (iv) cannabidiol (31 $\mu\text{g/kg}$). Ten minutes later, these subgroups received an i.v. continuous infusion of LPI (10 $\mu\text{g/kg min}$); after 10 min, a D–R curve to noradrenaline was again constructed during the infusion of LPI. Moreover, the fifth subgroup received an i. v. bolus injection of cannabidiol (31 $\mu\text{g/kg}$); 20 min later (with no i.v. infusion), a D–R curve to noradrenaline was again constructed to analyse the effect per se of this compound.

2.4. Drugs

Apart from the anaesthetic (diethyl ether), the compounds used in this study (obtained from the sources indicated) were: gallamine triethiodide, desipramine hydrochloride, (–)-noradrenaline bitartrate, anandamide, L- α -lysophosphatidylinositol, capsazepine, JWH-015 ([2-methyl-1-propyl-1H-indol-3-yl]-1-naphthalenylmethanone), NIDA41020 (1-(2,4-dichlorophenyl)-5-(4-methoxyphenyl)-4-methyl-N-(1-piperidinyl)-1H-pyrazole-3-carboxamide) (Sigma Chemical Co., St. Louis, MO, U.S.A.); AM630 (6-iodo-2-methyl-1-[2-(4-morpholinyl)ethyl]-1H-indol-3-yl)(4-methoxyphenyl)methanone) (Santa Cruz Biotechnology Inc., CA, U.S.A.); and cannabidiol (TOCRIS, Avomouth, Bristol, U.K.).

The compounds were dissolved in DMSO 0.5%, except: (i) gallamine, desipramine and noradrenaline (physiological saline); (ii) capsazepine and cannabidiol (DMSO 2%); and (iii) LPI (0.04 ml chloroform:methanol:water 70:27:3 + 0.96 ml saline). The resulting solutions were gauged with physiological saline. In the case of noradrenaline, ascorbic acid 1% was used to prevent oxidation. These vehicles had no effect on baseline diastolic blood pressure or heart rate (not shown). Fresh solutions were prepared for each experiment. The doses mentioned in the text refer to the free base of substances, except in the case of gallamine and desipramine, where they refer to their corresponding salts.

2.5. Data presentation and statistical analysis

All data in the text and figures are presented as mean $\pm$ S.E.M. The peak changes in diastolic blood pressure produced by either electrical stimulation or exogenous noradrenaline in saline- and agonist-infused animals were determined. The difference between the changes in diastolic blood pressure within one subgroup of animals was evaluated by Student–Newman–Keuls' post-hoc test, once a two-way repeated measures analysis of variance (randomized block design) had revealed that the samples represented different populations (Steel and Torrie, 1980). Statistical significance was accepted at $P < 0.05$.

3. Results

3.1. Systemic haemodynamic variables

The baseline values of diastolic blood pressure (DBP) and heart rate in the 234 rats were 58 ± 4 mm Hg and 301 ± 6 beats/min, respectively. A transient increase of both variables (not shown) was induced in the group receiving i.v. desipramine (50 μ g/kg), but returned to baseline values ($P > 0.05$) after 5 min. The latter values in desipramine-pretreated rats were not significantly modified by the continuous infusions of: (i) DMSO 0.5% (control; two times; 0.02 ml/min); (ii) LPI vehicle (0.04 ml chloroform:methanol:water 70:27:3 + 0.96 ml saline); and (iii) JWH-015 (5.6–10 μ g/kg min). In contrast, the basal values of DBP were significantly decreased by the continuous infusions of: (i) anandamide (0.31–3.1 μ g/kg min); and (ii) LPI (10 μ g/kg min) (see Table 2). Furthermore, the DBP values (in the absence of agonists) were not modified ($P > 0.05$) after i.v. bolus injections of: (i) 31 and 100 μ g/kg NIDA41020; (ii) 180–310 μ g/kg AM630; (iii) 310 μ g/kg capsazepine; or (iv) 31 μ g/kg cannabidiol (data not shown).

3.2. Effect of the antagonists on the hypotensive effect induced during the i.v. continuous infusions of anandamide and LPI

As shown in Table 3, the hypotensive effect induced by anandamide (1 μ g/kg min): (i) remained unaltered after DMSO (0.5 and 2%), AM630 (180–310 μ g/kg) or capsazepine (310 μ g/kg); and (ii) was prevented by NIDA41020 (31 and 100 μ g/kg) or cannabidiol (31 μ g/kg). Furthermore, the hypotensive effect induced by LPI was blocked by NIDA41020 (100 μ g/kg) or cannabidiol (10 and 31 μ g/kg), but not by the vehicle (see Table 3).

3.3. Initial effects produced by electrical sympathetic stimulation or exogenous noradrenaline on diastolic blood pressure and heart rate

The vasopressor responses induced by electrical stimulation of the vasopressor sympathetic outflow (0.03–3 Hz) or by exogenous noradrenaline (0.03–3 μ g/kg) were immediate and resulted in frequency- or dose-dependent increases in DBP (see control responses in Figs. 1–6). Exogenous noradrenaline also produced dose-dependent and transient increases in heart rate, as previously described (Villalón et al., 1995, 1998). The vasopressor responses to either electrical stimulation or noradrenaline were significant when compared with their baseline values (not shown). The electrically-induced vasopressor responses are induced by selective sympathetic stimulation, as

Table 2

Effect of the different i.v. continuous infusions of compounds ($n = 6$ each) on basal values of diastolic blood pressure (DBP) determined 10 min after a stable condition with no treatment (basal) or 10 min after each corresponding infusion had commenced.

Treatment	DBP (mm Hg)		
	Basal	After 1st dose treatment	After 2nd dose treatment
DMSO 0.5% (0.02 ml/min; given twice)	54 ± 5	52 ± 5	51 ± 4
LPI vehicle (0.02 ml/min; given twice)	59 ± 6	58 ± 4	54 ± 5
Anandamide (0.1 and 0.31 μ g/kg min)	55 ± 7	51 ± 4	43 ± 3^a
Anandamide (1 and 3.1 μ g/kg min)	54 ± 7	46 ± 6^a	44 ± 4^a
JWH-015 (1 and 10 μ g/kg min)	55 ± 5	56 ± 5	51 ± 5
LPI (1 and 3.1 μ g/kg min)	61 ± 3	62 ± 3	60 ± 2
LPI (5.6 and 10 μ g/kg min)	58 ± 7	54 ± 8	47 ± 8^a

DMSO, dimethyl-sulfoxide; LPI vehicle (0.04 ml chloroform:methanol:water 70:27:3 + 0.96 ml of saline); LPI, lysophosphatidylinositol.

^a $P < 0.05$ vs. the corresponding basal value. All values are expressed as mean $\pm$ S.E.M.

Table 3

Effect of the different i.v. bolus injections of compounds (i.e. before starting the infusion of either anandamide or LPI) on the hypotension induced by 1 μ g/kg min anandamide or 10 μ g/kg min LPI.

Pre-treatment	DBP (mm Hg)			
	Basal	Anandamide	Basal	LPI
DMSO 0.5% (1 ml/kg)	51 ± 5	41 ± 4^a	—	—
DMSO 2% (1 ml/kg)	57 ± 6	43 ± 2^a	64 ± 3	51 ± 2^a
NIDA41020 (31 μ g/kg)	63 ± 7	56 ± 5^a	—	—
NIDA41020 (100 μ g/kg)	68 ± 1	66 ± 2	64 ± 5	55 ± 4
AM630 (180 μ g/kg)	61 ± 3	48 ± 5^a	—	—
AM630 (310 μ g/kg)	58 ± 6	45 ± 5^a	—	—
Capsazepine (310 μ g/kg)	60 ± 5	48 ± 6^a	—	—
Cannabidiol (10 μ g/kg)	N.D.	N.D.	65 ± 4	60 ± 5
Cannabidiol (31 μ g/kg)	55 ± 7	46 ± 6	66 ± 4	66 ± 3

The basal values shown in this table correspond to those obtained 10 min after administering the above compounds. These values did not significantly differ ($P > 0.05$) from the corresponding control values (not shown for the sake of clarity).

^a $P < 0.05$ vs. the corresponding basal value. All values are expressed as mean $\pm$ S.E.M.

only negligible changes in heart rate were observed, as previously reported by our group (Manrique-Maldonado et al., 2011).

3.4. Effect of i.v. continuous infusions of vehicle, anandamide or LPI on the electrically-induced vasopressor responses with and without desipramine

Fig. 1 shows the vasopressor responses induced by electrical stimulation before (control) and during the i.v. continuous infusion of DMSO 0.5% (0.02 ml/min; 2 times), LPI vehicle (0.02 ml/min; 2 times), anandamide (1 and 3.1 μ g/kg min) and LPI (5.6 and 10 μ g/kg min) in rats with (upper panel) and without (lower panel) desipramine. In animals infused with vehicles, the electrically-induced vasopressor responses remained essentially unchanged when repeating two subsequent S–R curves (Fig. 1A, B, E and F). In contrast, the infusion of anandamide or LPI produced a significant inhibition of the electrically-induced vasopressor responses (Fig. 1C, D, G and H), with this inhibition being less prominent (but still significant) in the rats without desipramine.

3.5. Effect of vehicles, anandamide, JWH-015 and LPI on the vasopressor responses induced by either sympathetic stimulation or exogenous noradrenaline

Fig. 2 shows the vasopressor responses induced by electrical sympathetic stimulation before (control) and during i.v. continuous infusions of vehicles or the agonists. Accordingly, in animals infused with vehicles (0.02 ml/min), the electrically-induced vasopressor responses remained unchanged when repeating two subsequent S–R curves (Fig. 2A and B). Similar results (i.e. no inhibition) were obtained during the infusions of JWH-015 (5.6–10 μ g/kg min; Fig. 2F). In contrast, the infusions of 0.1 and 0.31 μ g/kg min anandamide dose-dependently inhibited the vasopressor responses induced by 0.3, 1 and 3 Hz (Fig. 2C), while 1 and 3.1 μ g/kg min anandamide produced the same degree of inhibition (at 0.3, 1 and 3 Hz, $P > 0.05$; see Fig. 2D). Likewise, the infusions of 5.6 and 10 μ g/kg min LPI produced a dose-dependent inhibition of the vasopressor responses induced by 0.3, 1 and 3.1 Hz (Fig. 2E), while 1 and 3.1 μ g/kg min LPI produced no inhibition (not shown). For these reasons, 1 μ g/kg min anandamide and 10 μ g/kg min LPI were chosen for further pharmacological analysis in rats with desipramine (see below).

Moreover, Fig. 3 shows the vasopressor responses induced by i.v. bolus injections of noradrenaline before (control) and during i.v. continuous infusions of vehicles or the agonists. Hence, in animals infused with vehicles (0.02 ml/min), the noradrenaline-induced

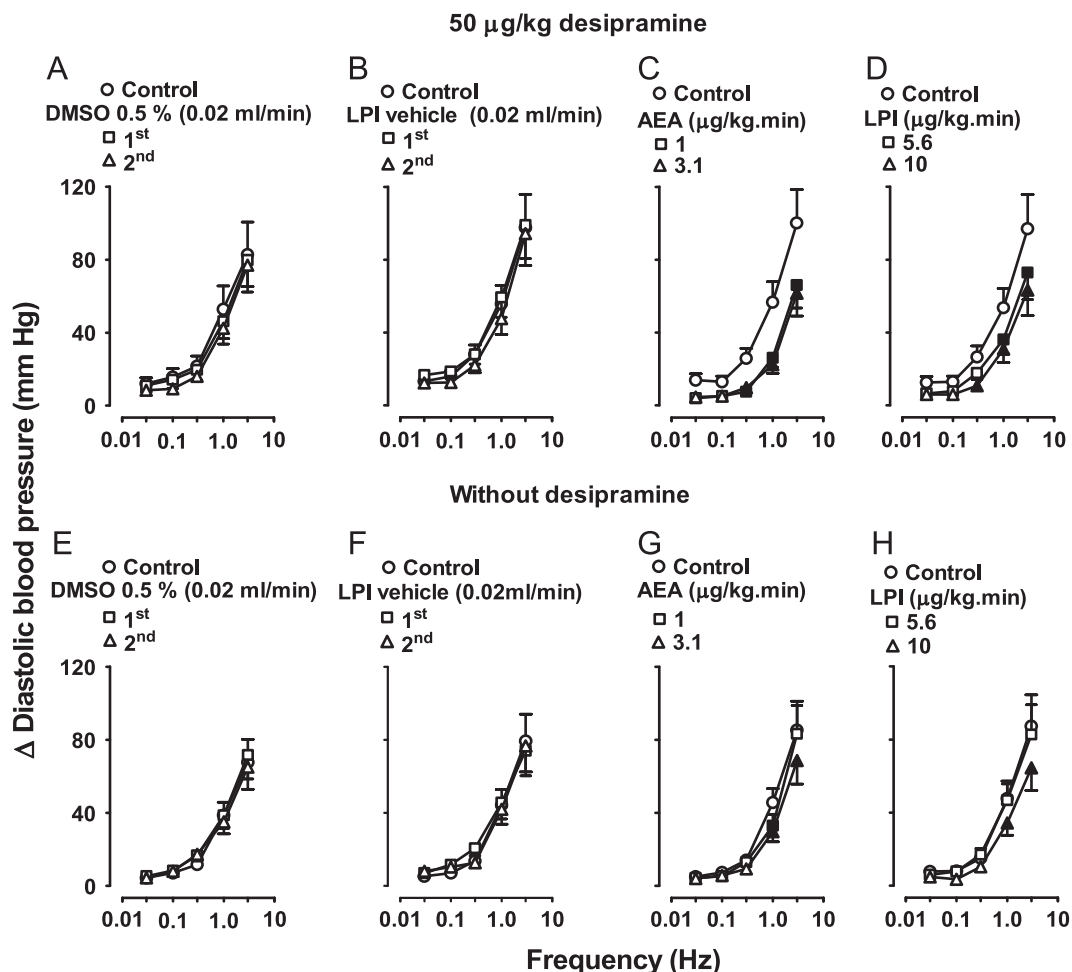


Fig. 1. Effect of an i.v. infusion ($n=6$ each figure) of (A and E) DMSO 0.5% (0.02 ml/min; twice); (B and F) vehicle of lysophosphatidylinositol (LPI vehicle: 0.04 ml chloroform: methanol:water 70:27:3+0.96 ml saline; 0.02 ml/min, 2 times); (C and G) anandamide (AEA; 1 and 3.1 $\mu\text{g/kg min}$); and (D and H) LPI (5.6 and 10 $\mu\text{g/kg min}$) in the presence (upper panel) or in the absence (lower panel) of desipramine (50 $\mu\text{g/kg}$, i.v.) on the electrically-induced vasopressor responses in pithed rats. For the sake of clarity, empty symbols depict either control responses ($\circ$) or non-significant ($P > 0.05$) responses (Δ) vs. control. Solid symbols ($\bullet$) represent significantly different responses ($P < 0.05$) vs. control.

vasopressor responses remained unchanged when repeating two subsequent D-R curves (Fig. 3A and B). Similar results (i.e. no inhibition) were obtained during the infusions of anandamide (1 and 3.1 $\mu\text{g/kg min}$; Fig. 3C). Interestingly, the infusions of 5.6 and 10 $\mu\text{g/kg min}$ LPI produced a dose-dependent inhibition of the noradrenaline-induced vasopressor responses (Fig. 3D), as previously observed with the electrically-induced vasopressor responses (see above; Fig. 2E).

3.6. Effect of vehicles, NIDA41020, AM630, capsazepine or cannabidiol per se on the vasopressor responses induced electrically or by i.v. bolus injections of noradrenaline

Fig. 4 shows that the electrically-induced vasopressor responses remained unchanged (as compared with the control S-R curves; $P > 0.05$) after an i.v. bolus injection of: (i) 1 ml/kg DMSO (0.5% and 2%; Fig. 4A and B, respectively); (ii) 100 $\mu\text{g/kg}$ NIDA41020 (Fig. 4C); (iii) 180 $\mu\text{g/kg}$ AM630 (Fig. 4D); (iv) 310 $\mu\text{g/kg}$ AM630 (Fig. 4E); (v) 310 $\mu\text{g/kg}$ capsazepine (Fig. 4F); and (v) 31 $\mu\text{g/kg}$ cannabidiol (Fig. 4G). Likewise, Fig. 4H shows that the noradrenaline-induced vasopressor responses were not affected (as compared with the control D-R curves; $P > 0.05$) after 31 $\mu\text{g/kg}$ cannabidiol. It is to be noted that, with the exception of 31 $\mu\text{g/kg}$ cannabidiol, the rest of the above antagonists were not tested on the D-R curves to noradrenaline as anandamide failed to inhibit these D-R curves (see Fig. 3C).

3.7. Effect of vehicle, NIDA41020 or cannabidiol on LPI-induced inhibition of the vasopressor responses induced by electrical stimulation or exogenous noradrenaline

Fig. 5 (upper panel) shows that the inhibition by 10 $\mu\text{g/kg min}$ LPI on the electrically-induced vasopressor responses was: (i) unaltered after 1 ml/kg DMSO 2% (Fig. 5A); (ii) slightly blocked (only at 0.3 Hz) by 100 $\mu\text{g/kg}$ NIDA41020 (Fig. 5B); and (iii) markedly blocked (Fig. 5C) and abolished (Fig. 5D) by 10 and 31 $\mu\text{g/kg}$ cannabidiol, respectively. Likewise, Fig. 5 (lower panel) depicts that the inhibition by 10 $\mu\text{g/kg min}$ LPI on the noradrenaline-induced vasopressor responses was (i) unaltered after 1 ml/kg DMSO 2% (Fig. 5E); (ii) slightly blocked (only at 0.3 $\mu\text{g/kg}$ noradrenaline) by 100 $\mu\text{g/kg}$ NIDA41020 (Fig. 5F); and (iii) dose-dependently blocked by 10 and 31 $\mu\text{g/kg}$ cannabidiol, with complete blockade at its highest dose (Fig. 5G and H, respectively).

3.8. Effect of i.v. bolus injections of vehicles, NIDA41020, AM630, capsazepine or cannabidiol on anandamide-induced inhibition of the vasopressor sympathetic outflow

Fig. 6 shows that the sympatho-inhibition by 1 $\mu\text{g/kg min}$ anandamide was: (i) unaltered after 1 ml/kg DMSO 0.5% (Fig. 6A), 1 ml/kg DMSO 2% (Fig. 6B), 180 $\mu\text{g/kg}$ AM630 (Fig. 6E) or 310 $\mu\text{g/kg}$ capsazepine (Fig. 6G); (ii) markedly blocked and abolished by 31 and 100 $\mu\text{g/kg}$

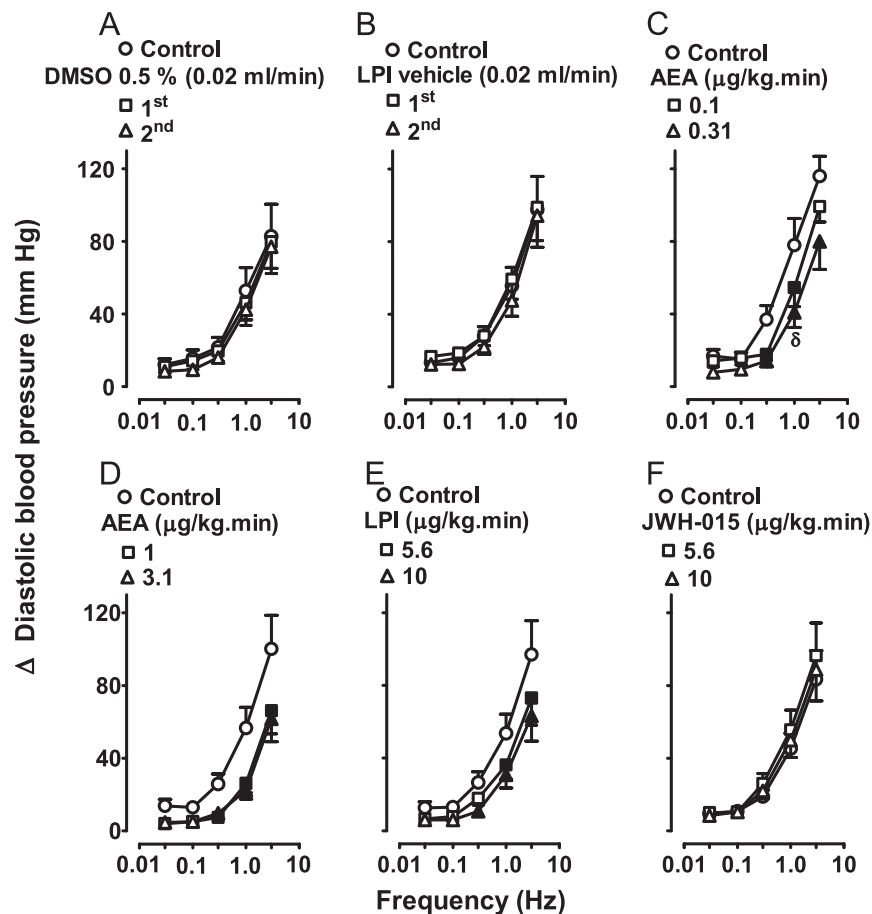


Fig. 2. Effect of i.v. continuous infusions ($n=6$ each figure) of (A) vehicle of both anandamide and JWH-015 (i.e. DMSO 0.5%; 0.02 ml/min); (B) vehicle of lysophosphatidylinositol (LPI vehicle: 0.04 ml chloroform:methanol:water 70:27:3+0.96 ml saline; 0.02 ml/min); (C and D) anandamide (AEA; 0.1–3.1 μ g/kg min); (E) LPI (5.6 and 10 μ g/kg min); or (F) JWH-015 (5.6–10 μ g/kg min) on the electrically-induced vasopressor responses. For the sake of clarity, empty symbols depict either control responses ($\circ$) or non-significant ($P > 0.05$) responses (Δ) vs. control. Solid symbols ($\bullet$) represent significantly different responses ($P < 0.05$) vs. control. Note that (A), (B), (D) and (E) are the same as those shown in Figs. 1A, B, C and D, respectively. $^*P < 0.05$ when comparing the inhibition by 0.1 μ g/kg min AEA vs. that by 0.31 μ g/kg min AEA in (C).

kg NIDA41020, respectively (Fig. 6C and D); and (iii) slightly blocked by 310 μ g/kg AM630 (Fig. 6F) or 31 μ g/kg cannabidiol (Fig. 6H).

4. Discussion

4.1. General

Our results show that the sympatho-inhibition by 1 μ g/kg min anandamide and the inhibition by 10 μ g/kg min LPI of the vasopressor responses to electrical stimulation and exogenous noradrenaline are operative in the absence of desipramine (compare upper and lower panels of Fig. 1). As expected, the inhibition by both compounds was less prominent without desipramine; however, desipramine was systematically used as the inhibition by anandamide and LPI was more pronounced at lower stimulation frequencies. This facilitated the pharmacological identification of the mechanisms involved (Figs. 5,6), as previously discussed for other substances (Villalón et al., 1995; 1998). Since anandamide inhibited the sympathetically-induced vasopressor responses (Fig. 2C and D), but not those by noradrenaline (Fig. 3C), the role of sympatho-inhibitory (rather than postjunctional) mechanisms is reinforced. In contrast, the fact that LPI similarly inhibited the vasopressor responses induced by both procedures (Figs. 2E and 3D) supports the role of postjunctional (vascular), rather than sympatho-inhibitory, mechanisms.

Anandamide-induced sympatho-inhibition is mainly mediated by cannabinoid CB₁ receptors (abolished by 100 μ g/kg NIDA41020; Fig. 6D) and, to a much lesser extent (only at 0.3 Hz), by a cannabidiol-sensitive mechanism (Fig. 6H). Based on cannabidiol binding properties (Table 1), this mechanism may involve GPR55 and/or cannabinoid CB₁ receptors. Clearly, the characterization of this cannabidiol-sensitive component (Fig. 6H) would require additional experiments, which fall beyond the scope of this study; the selective GPR55 antagonist CID16020046 (Kargl et al., 2013) would be a valuable tool.

The sympatho-inhibitory role of cannabinoid CB₁ receptors is consistent with its coupling to G_{i/o}-proteins, which results in inhibition of adenylyl cyclase, inactivation of Ca²⁺ channels and/or activation of the inwardly rectifying K⁺ channels (Sugiura and Waku, 2002); these are signal transduction systems associated with inhibition on the release of neuromediators (Boehm and Kubista 2002; De Jong and Verhage 2009).

The LPI-induced inhibition of the vasopressor responses produced electrically (Fig. 2E) and by noradrenaline (Fig. 3D) is consistent with the possible involvement of receptors that (i) resemble the pharmacological profile of GPR55 (blocked by cannabidiol; see Fig. 5C and D, 5G and H); and (ii) are located on vascular endothelium (Daly et al., 2010) and whose signal transduction system involves an increase in intracellular Ca²⁺ (Henstridge et al., 2009), although we have no direct evidence that these mechanisms are operative in our experimental model. On the other hand, the fact that 100 μ g/kg NIDA41020, a

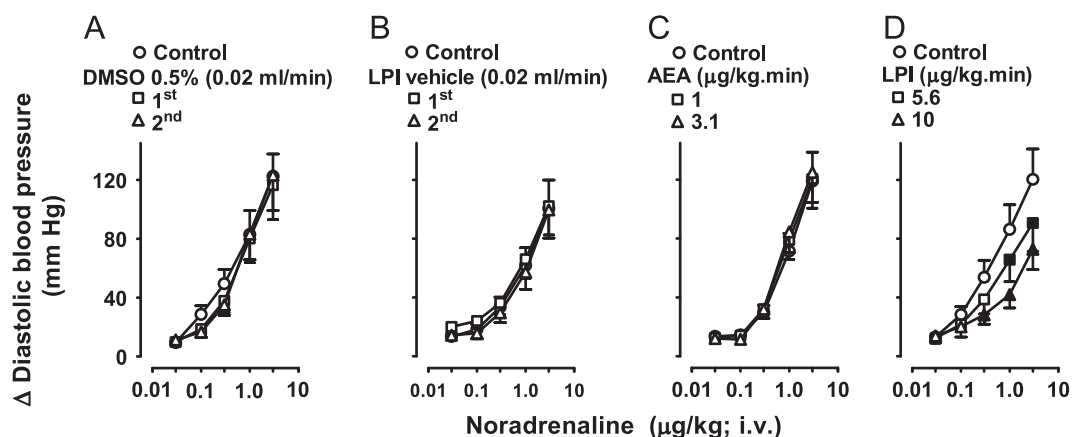


Fig. 3. Effect of i.v. continuous infusions ($n=6$ each figure) of: (A) vehicle of anandamide (DMSO 0.5%; 0.02 ml/min); (B) vehicle of lysophosphatidylinositol (LPI vehicle; 0.04 ml chloroform:methanol:water 70:27:3 + 0.96 ml saline; 0.02 ml/min); (C) anandamide (AEA; 1 and 3.1 $\mu\text{g/kg min}$); or (D) LPI (5.6 and 10 $\mu\text{g/kg min}$) on the vasopressor responses induced by i.v. bolus injections of exogenous noradrenaline in pithed rats. For the sake of clarity, empty symbols depict either control responses ($\circ$) or non-significant ($P > 0.05$) responses (Δ) vs. control. Solid symbols ($\blacksquare$) represent significantly different responses ($P < 0.05$) vs. control. $^a P < 0.05$ when comparing the inhibition by 5.6 $\mu\text{g/kg min}$ LPI vs. that by 10 $\mu\text{g/kg min}$ LPI in (D).

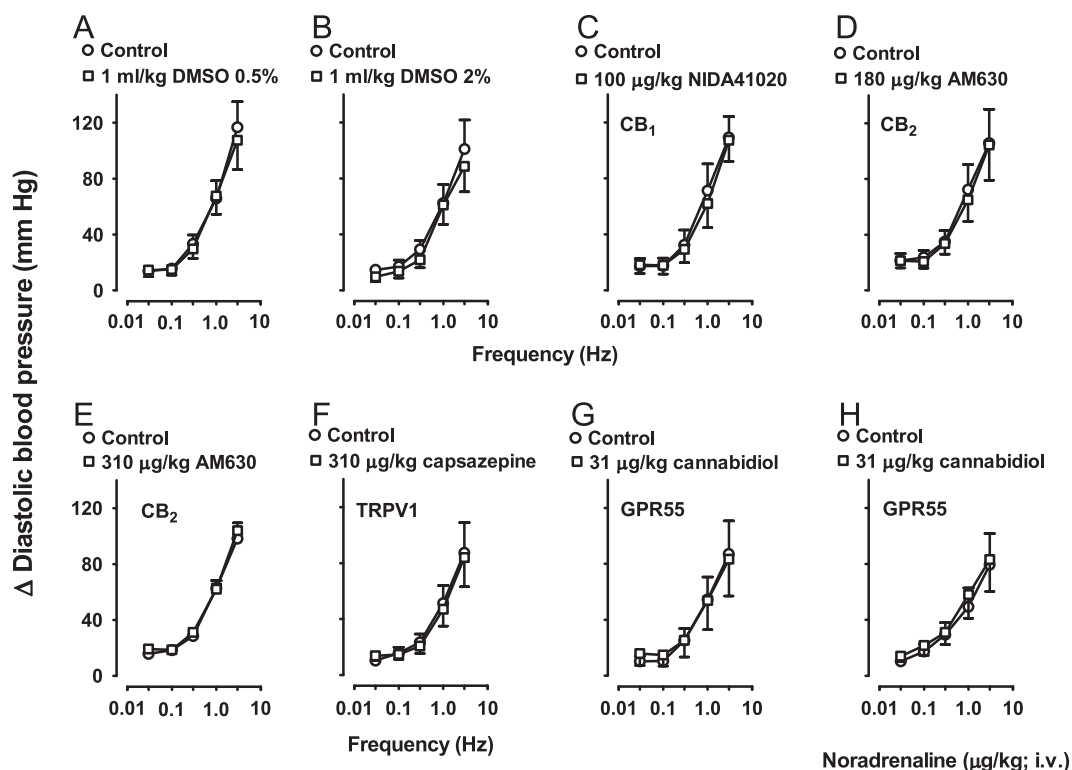


Fig. 4. Effect per se of i.v. bolus injections ($n=6$ each figure) of: (A) 1 ml/kg DMSO 0.5%; (B) 1 ml/kg DMSO 2%; (C) 100 $\mu\text{g/kg}$ NIDA41020; (D, E) 180 and 310 $\mu\text{g/kg}$ AM630; (F) 310 $\mu\text{g/kg}$ capsazepine; or (G, H) 31 $\mu\text{g/kg}$ cannabidiol on the electrically-induced vasopressor responses; or (H) 31 $\mu\text{g/kg}$ cannabidiol on the noradrenaline-induced vasopressor responses in pithed rats. For the sake of clarity, empty symbols depict either control responses ($\circ$) or non-significant ($P > 0.05$) responses ($\square$) vs. control.

cannabinoid CB_1 receptor antagonist (Table 1), minimally blocked LPI-induced inhibition of the vasopressor responses produced electrically (at 0.3 Hz; Fig. 5B) and by noradrenaline (at 0.3 $\mu\text{g/kg}$; Fig. 5F) may suggest (i) a small participation (if any) of cannabinoid CB_1 receptors in the above inhibition by LPI; and/or (ii) a slight blockade of GPR55 by NIDA41020, although its affinity for these receptors is unknown (Table 1). Notwithstanding, since LPI displays about a 3 log unit higher affinity for GPR55 as compared to cannabinoid CB_1 receptors (Table 1), the second suggestion seems most likely.

4.2. Systemic haemodynamic changes

The inhibition by anandamide and LPI of the vasopressor responses by electrical stimulation and/or exogenous noradrenaline (Figs. 1–3, 5, and 6) cannot be explained in terms of: (i) tachyphylaxis, as the corresponding S–R and D–R curves remained unchanged after i.v. continuous infusions of DMSO 0.5% (Fig. 2A) or LPI vehicle (Fig. 2B); and (ii) time-dependent experimental changes. Moreover, the inhibition by anandamide or LPI of the vasopressor responses could have been due to the

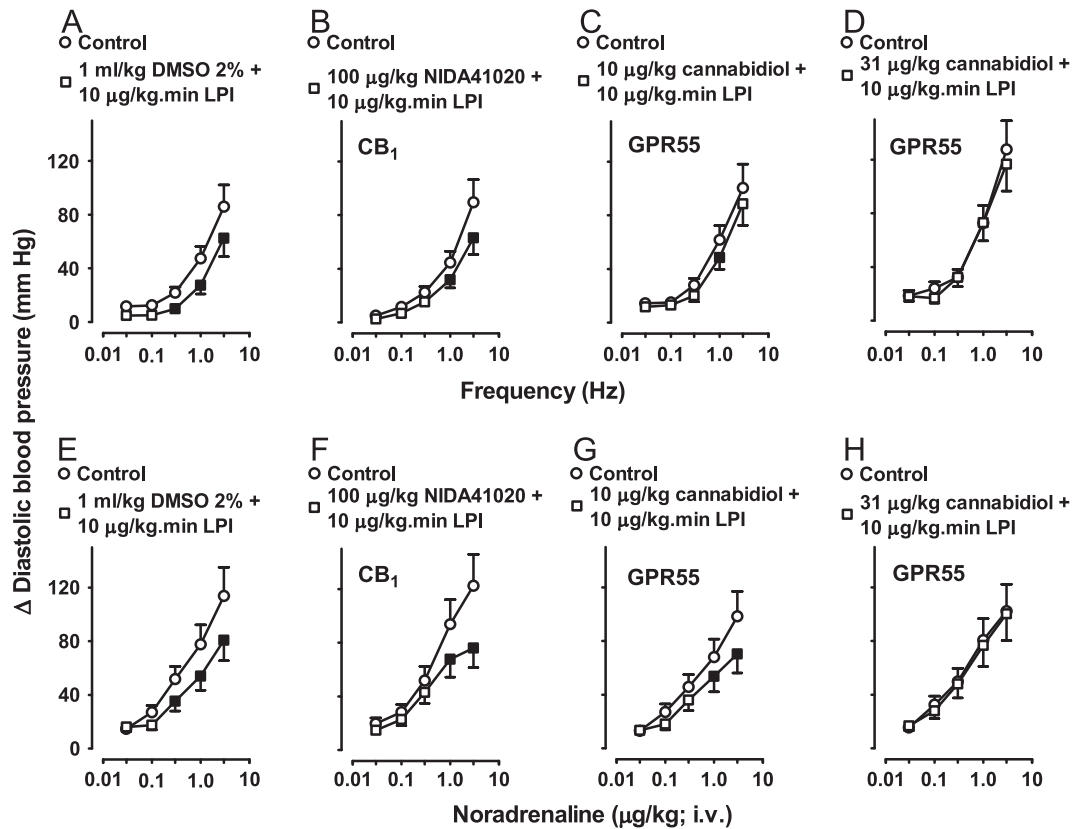


Fig. 5. Effect of i.v. bolus injections of: (A, E) 1 ml/kg DMSO 2%; (B and F) 100 μ g/kg NIDA41020; (C and G) 10 μ g/kg cannabidiol; and (D and H) 31 μ g/kg cannabidiol on lysophosphatidylinositol (LPI)-induced inhibition of the vasopressor responses induced by either electrical stimulation (upper panel) or i.v. bolus injections of exogenous noradrenaline (lower panel) in pithed rats. For the sake of clarity, empty symbols depict either control responses ($\circ$) or non-significant ($P > 0.05$) responses ($\square$) vs. control. Solid symbols ($\blacksquare$) represent significantly different responses ($P < 0.05$) vs. control.

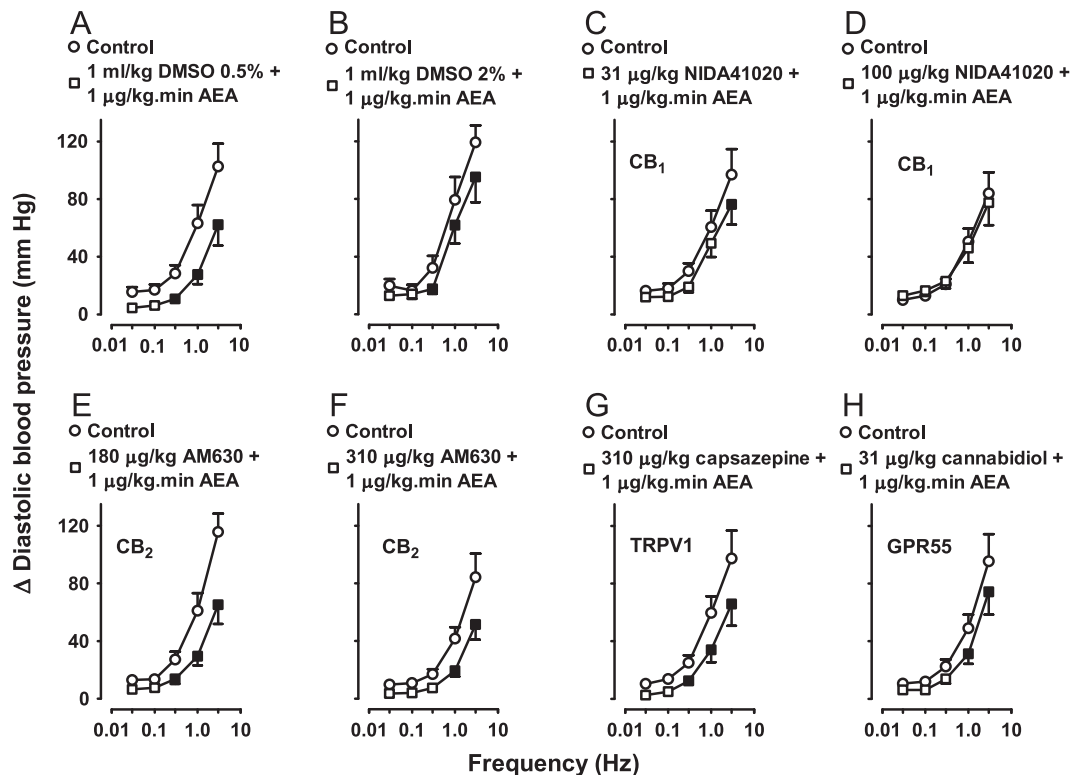


Fig. 6. Effect of i.v. bolus injections of: (A, B) DMSO 0.5 and 2% (1 ml/kg each); (C, D) 31 and 100 μ g/kg NIDA41020; (E, F) 180 and 310 μ g/kg AM630; (G) 310 μ g/kg capsazepine; or (H) 31 μ g/kg cannabidiol on anandamide (AEA)-induced inhibition of the vasopressor sympathetic outflow in pithed rats. For the sake of clarity, empty symbols depict either control responses ($\circ$) or non-significant ($P > 0.05$) responses ($\square$) vs. control. Solid symbols ($\blacksquare$) represent significantly different responses ($P < 0.05$) vs. control.

hypotension produced by both compounds (Table 2); however, the lowest doses of NIDA41020 and cannabidiol, which abolished this hypotension (Table 3), failed to abolish LPI- (Fig. 5B and C) or anandamide-induced (Fig. 6C and H) inhibition. The above hypotension reflects peripheral mechanisms as pithed rats were used. Interestingly, Zakrzewska et al. (2010) have reported that an i.v. bolus injection of anandamide produces a transient vasopressor response (phase II) followed by a late vasodepressor response (phase III) in pithed rats. This vasopressor response was not seen in our study probably because we used i.v. continuous infusions of anandamide. Thus, the anandamide-induced hypotension in our study (Table 2) most likely: (i) reflects the phase III of the triphasic blood pressure response to anandamide (Zakrzewska et al., 2010); and (ii) involves cannabinoid CB₁ and cannabidiol-sensitive (probably GPR55) receptors (Table 1) based on the blockade produced by 31 and 100 µg/kg NIDA41020 or 31 µg/kg cannabidiol (see Tables 1 and 3). Although our experiments do not allow us to establish whether these receptors are located on vascular endothelium and/or vascular smooth muscle, other studies have described that cannabinoid CB₁ receptors are located at both vascular levels (Liu et al., 2000; Sugiura et al., 1998); whereas GPR55 is located on endothelial cells (Bondarenko et al., 2010; 2011). Indeed, LPI-induced hypotension (Table 2) suggests the possible role of GPR55 producing vasorelaxation as LPI can activate GPR55-dependent and independent mechanisms on vascular endothelium (Bondarenko et al., 2010; 2011). Hence, the blockade by 100 µg/kg NIDA41020 of LPI-induced hypotension (Table 3) could suggest: (i) the role of cannabinoid CB₁ receptors; and/or (ii) an antagonistic activity of NIDA41020 on GPR55.

Since the antagonists NIDA41020 (CB₁), AM630 (CB₂), capsaizipine (TRPV1) or cannabidiol (GPR55) did not affect baseline blood pressure (not shown), heart rate (not shown) or the vasopressor responses (Fig. 4): (i) their respective receptors do not seem to play a role in the modulation of systemic vascular tone; (ii) these compounds, at the doses used, are devoid of any effect per se on the vasopressor responses.

4.3. Pharmacological analysis of the sympatho-inhibition by anandamide

The fact that anandamide-induced sympatho-inhibition remained unchanged after DMSO (0.5% and 2%; compare Fig. 6A and B with Fig. 2D) indicates that this response is reproducible. Moreover, although the possible interference by pharmacokinetic factors in pithed rats cannot be excluded, anandamide-induced sympatho-inhibition: (i) most likely involves cannabinoid CB₁ receptors as it was markedly blocked and abolished by 31 and 100 µg/kg NIDA41020 (Fig. 6C and D, respectively); (ii) does not involve cannabinoid CB₂ or vanilloid TRPV1 receptors as it remained unaffected after 180 µg/kg AM630 (Fig. 6E) or 310 µg/kg capsaizipine (Fig. 6G); and (iii) involves, to a lesser extent, a cannabidiol-sensitive receptor slightly blocked by 31 µg/kg cannabidiol (only at 0.3 Hz; Fig. 6H), a dose that abolished LPI-induced inhibition (Fig. 5C and F). Nevertheless, cannabidiol can interact with GPR55 and cannabinoid CB₁ receptors (Table 1). Interestingly, the slight blockade produced by 310 µg/kg AM630 on anandamide-induced inhibition at 0.3 Hz (Fig. 6F) apparently supports a minor role, if any, of cannabinoid CB₂ receptors. Notwithstanding: (i) the fact that AM630 displays also a weak affinity for cannabinoid CB₁ receptors (Table 1) agrees with the role of these receptors as suggested by the blockade by NIDA41020 (see above); and (ii) the failure of JWH-015 (a cannabinoid CB₂ receptor agonist; Table 1) to induce sympatho-inhibition (Fig. 2F) excludes the role of cannabinoid CB₂ receptors. It must be emphasized that, on the basis of their affinities (Table 1), the doses used of NIDA41020, AM630, capsaizipine and cannabidiol were high enough to block their respective receptors (see Figs. 5 and 6). Similarly, the

role of vanilloid TRPV1 receptors (see Section 1) can be excluded since their signal transduction system (i.e. an increase in Na⁺/Ca²⁺ channel conductance) is associated with facilitation (rather than inhibition) of transmitter release (Poblete et al., 2005).

Regarding the possible locus of the cannabinoid CB₁ receptors involved in anandamide-induced sympatho-inhibition, central mechanisms are not operative in our experimental model since pithed rats were used. However, we cannot exclude an inhibitory action of anandamide at both the autonomic ganglia and post-ganglionic sympathetic neurons, which have modulatory cannabinoid CB₁ receptors (Ishac et al., 1996).

4.4. Evidence supporting the possible role of GPR55 in the inhibition by LPI

LPI-induced inhibition of the vasopressor responses produced by electrical stimulation and noradrenaline is reproducible as it remained unchanged after DMSO 2% (compare Fig. 5A with Fig. 2E; and Fig. 5E with Fig. 3D). Hence, since both responses were markedly blocked by 10 µg/kg cannabidiol (Fig. 5C and G) and abolished by 31 µg/kg cannabidiol (Fig. 5D and H), but only minimally blocked by 100 µg/kg NIDA41020 (Figs. 5B and F), a possible (but not exclusive) role of GPR55 is suggested. Indeed, these receptors: (i) produce vasorelaxation (Yousuf et al., 2010); and (ii) are located on vascular endothelium (Bondarenko et al., 2010; 2011). Likewise, Waldeck-Weiermair et al. (2008) showed that both GPR55 and cannabinoid CB₁ receptors: (i) are functional on vascular endothelium; and (ii) can be activated by anandamide with different transduction mechanisms depending on the state of integrins. Hence, the inactivity of anandamide on noradrenaline-induced vasopressor responses (Fig. 3C) excludes post-junctional vasorelaxing mechanisms. In contrast, the inhibition of noradrenaline-induced vasopressor responses by LPI (Fig. 3D) suggests the role of postjunctional vasorelaxing mechanisms (probably via endothelial GPR55 receptors), as reported by Yousuf et al. (2010).

5. Conclusion

Our results, taken together, suggest that anandamide-induced inhibition of the vasopressor sympathetic outflow is predominantly mediated by sympatho-inhibitory cannabinoid CB₁ and, to a lesser extent (if different from CB₁), by cannabidiol-sensitive receptors (presumably GPR55), with no evidence for the involvement of cannabinoid CB₂ and vanilloid TRPV1 receptors. In contrast, LPI-induced inhibition of the vasopressor responses produced by electrical stimulation and exogenous noradrenaline mainly involves postjunctional (rather than sympatho-inhibitory) cannabidiol-sensitive (probably GPR55) receptors mediating vasorelaxation. These receptors are most likely located on vascular endothelium, as previously suggested (Bondarenko et al., 2010; 2011; Yousuf et al., 2010).

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